AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

Chemistry Paper I

Time allowed: 35 minutes Marks 25

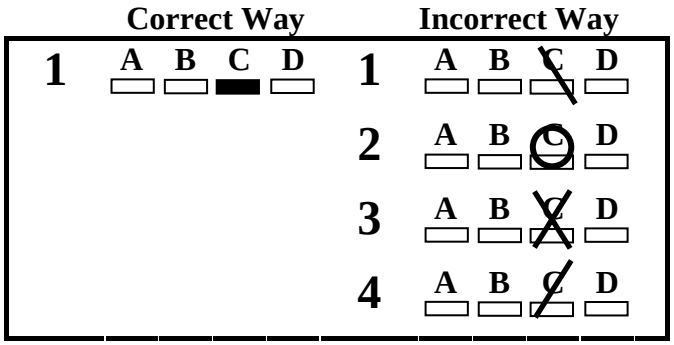
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. Ingredients of baking powder are listed below.

Baking Powder Ingredients

Corn starch =4gms
Sod. Bicarbonate = 0.2gms
Calcium acid phosphate =0.01gms
Sod. Aluminum Sulphate = 0.09gms

Baking powder is classified as a/an

A. amalgam.
B. compound.
C. heterogenous mixture.
D. homogenous mixture.

2. Which type of chemical reaction occurs when zinc metal reacts with dilute sulphuric acid?

A. Addition reaction
B. Decomposition reaction
C. Displacement reaction
D. Double displacement reaction

3. A charged group of atoms which behaves as a single unit in a chemical reaction is known as a

A. radical.
B. molecule.
C. complex molecule.
D. formula unit.

4. The mass of an atom of an element depends upon the number of its

A. protons and electrons.
B. protons and neutrons.
C. electrons and neutrons.
D. protons, neutrons and electrons.

5. How many electrons are present in the outermost shell of chlorine? (atomic.mass of Cl=35)

A. 5
B. 6
C. 7
D. 8

6. A local scientist discovered an element and named it Pakistanium (Pk) which contains 111 protons and 141 neutrons.

Which of the following is the correct representation of an atom of Pakistanium?

A. ${}_{30}\mathrm{Pk}^{141}$
B. ${}_{30}\mathrm{Pk}^{111}$
C. ${}_{111}\mathrm{Pk}^{141}$
D. ${}_{111}\mathrm{Pk}^{252}$

7. How many orbitals are present in the p-subshell of an atom?

A. 1
B. 2
C. 3
D. 4

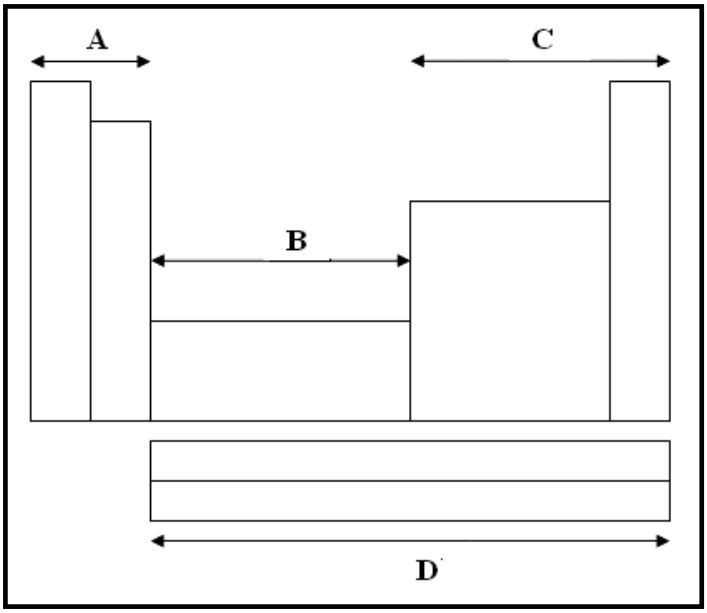
8. The elements in the modern periodic table are arranged in order of their

A. atomic size.
B. atomic number.
C. atomic mass.
D. date of discovery.

9. Which of the following properties decreases in a period from left to right in a periodic table?

A. Atomic radii
B. Nuclear charge
C. Electronegativity
D. Ionization energy

10. Identify the d-block elements in the given periodic table.



11. During the formation of an ionic bond, atoms

A. gain protons.
B. gain electrons.
C. lose electrons.
D. gain and lose electrons.

12. Which of the following bonds is formed when a molecule donates an electron pair to another molecule?

A. Ionic bond
B. Covalent bond
C. Metallic bond
D. Co-ordinate covalent bond

13. Which of the following molecules contain polar covalent bonds?

A. Ammonia
B. Chlorine
C. Hydrogen
D. Oxygen

14. Which of the following is classified as a weak force of interaction?

A. Ionic bond
B. Covalent bond
C. Hydrogen bond
D. Metallic bond

15. Which of the following processes involves the increase of kinetic energy of molecules of a liquid by heating?

A. Condensation
B. Evaporation
C. Sublimation
D. Melting

16. All of the following are examples of amorphous solids EXCEPT

A. glass.
B. rubber.
C. plastic.
D. sulphur.

17. Which of the following types of solution is found in carbonated water?

A. Gas and solid
B. Gas and liquid
C. Liquid and solid
D. Liquid and liquid

18. All of the following are insoluble in water EXCEPT

A. benzene.
B. methanol.
C. chloroform.
D. kerosine oil.

19. How many moles of solute are present in $1 dm^3$ of 2 M solution?

A. 0 mole
B. 1 mole
C. 2 moles
D. 3 moles

20. A theory which explains the process of electrolysis is called

A. ionic.
B. current.
C. covalent.
D. co-ordinate covalent.

21. Name the species which gets reduced in the equation given below.

$$CuO + H_2 \rightarrow Cu + H_2O$$

A. Cu
B. CuO
C. H_2O
D. H_2

22. What products are formed when sodium metal reacts with water?

A. Sodium oxide and hydrogen
B. Sodium hydroxide and hydrogen
C. Sodium hydroxide and oxygen
D. Sodium hydride and oxygen

23. Which of the following have a low melting point?

A. Metals
B. Metalloids
C. Non-metals
D. Transition metals

24. When Ca and Mg burn in air they form

A. acidic oxide.
B. basic oxide.
C. neutral oxide.
D. amphoteric oxide.

25. The alkali metals are more reactive because of their

A. high ionization energy.
B. low ionization energy.
C. high electronegativity.
D. high electron affinity.

END OF PAPER